BOM requirements

What the schematic has to buy, and the simulation that says so.

Every row here exists because a block in `sim/blocks/` failed without it. The rule this file follows is the same one the netlists follow: **a figure nobody has read in a datasheet does not become a design number.** So these are requirements stated as specifiable, purchasable properties — tempco bands, ESR bands, on-resistance ceilings — not manufacturer part numbers. Naming a part is a separate step, and it needs a retrieved datasheet.

Where a part number *does* appear below, it is because it was checked.

Input protection

`TVS` — transient suppressor, battery input

SMCJ43CA. Bidirectional, 43 V standoff, 1500 W package.

Both halves of that were forced by simulation, not chosen:

- **43 V standoff** is the smallest standard bin above the 40 V clamped load dump. Below it the part conducts during a *normal* ISO 7637-2 pulse 5b event rather than a fault — which is what `load_dump.cir` was failing on, not "the TVS is too small."
- **1500 W (SMC) package, not 600 W (SMB).** At the SMB size, pulse 2a clamps at 88.6 V against the buck's 100 V absolute maximum — 11% margin. The SMC part has roughly half the dynamic resistance at the same standoff and clamps at 73.3 V, 1.37×.

Checked by `load_dump.cir` (0.000 A, 0.000 J through the TVS) and `transient_clamp.cir`.

`NEG-CLAMP` — negative active clamp, buck VIN

N-channel MOSFET, **Vds ≥ 80 V (100 V class), Rds(on) ≤ 10 mΩ at Vgs = 4.5 V**, source to ground, drain to VIN, driven by a comparator sensing VIN against a −0.1 V threshold. **In parallel with** the 100 V / 20 A clamp Schottky, which stays as the instant passive backstop.

> **CORRECTED 22 September 2026 — both parts were under-rated for the rail they sit on.** This tag originally read **Vds ≥ 40 V** and the backstop Schottky was **60 V**. Both have a terminal on `VBAT_PROT`, and both were sized for the *negative* pulses they exist to catch — but pulse 2a drives the **same node to +73.3 V**, where a 40 V FET is 0.55× rated and a 60 V Schottky 0.82×. The 40 V FET had zero margin on the 40 V load dump

> alone. `transient_clamp.cir` already carried the rule "every part on VBAT_PROT must clear 73.3 V" — as a principle, not a list, so nothing compared these two against it. It is now a table in `run_sim.py`: every part with a terminal on this rail, its rating, and the margin, and a part below the line fails.

The requirement is really an *effective clamp impedance*, and it is worth stating as one: the LM5164's VIN absolute maximum is −0.3 V, pulse 1 delivers 150 V through 10 Ω, so

```
Rclamp ≤ 0.3 V / 15 A = 20 mΩ, junction drop included
```

No junction device meets that, and the reason is structural rather than a sizing miss. Decomposing the Schottky's own drop at 15 A:

term	value
junction, `N·Vt·ln(I/Is)`	0.367 V
bulk, `I·Rs`	0.180 V
total	**0.547 V**

Only the bulk term falls when parts are paralleled; the junction term is logarithmic in current and barely moves. Four Schottkys in parallel still leave 0.375 V. A FET in its ohmic region has no junction offset at all — 15 A through 5 mΩ is 75 mV.

Checked by `negative_pulses.cir`: buck VIN reaches −0.068 V on pulse 1 (4.4× inside the rating) and −0.027 V on pulse 3a (11×), against −0.528 V and −0.393 V with the Schottky alone.

Closed 22 September 2026 — and closing it found a latch the block had been reporting as a pass.

The control is **TLV3201-Q1** (50 ns max, push-pull, no phase inversion beyond the rails) into **UCC27517A-Q1** (4 A, 23 ns max at 12 V, output held low in UVLO). The comparator's 52 mA would take about half a microsecond to charge the clamp FET's gate; the driver takes nanoseconds.

The clamp is a low-side ideal diode, and it has to let go. The model engaged it near VIN = +0.08 V and released it only above +0.12 V. At the end of every negative pulse the source returns to +13.5 V and drives current back through the still-on clamp — 1.35 A through 5 mΩ is 6.7 mV, never reaching the release threshold — so **the clamp latched on and shorted the source for the rest of the run.** `run_sim.py` checked only the clamp branch's *minimum* and reported it RESOLVED. A recovery check added the same day failed against it on both pulse 1 and pulse 3a. With a real battery instead of the test generator's 10 Ω, that is a dead short after the first negative pulse.

The thresholds are now **engage below −100 mV, release above −6.8 mV**, from the drawn network and checked as arithmetic against it:

sense node	10 kΩ from VIN, 499 kΩ from 5V_MAIN: 0.9804·VIN + 98.2 mV, so −100 mV arrives as 0 V and no negative reference is needed
reference	10 kΩ to ground, **536 kΩ** from the output: 0 V low, 91.6 mV high
release, worst case	−6.8 mV nominal, **−1.8 mV** with the comparator's full 5 mV offset — still below zero, so it cannot hold on against reverse current (523 kΩ, the first value, sat at +0.5 mV)

The window before it engages is now a number, and it is inside the rating. With the ~100 ns delay (50 + 23 ns max plus gate charge) and a 10 ns timestep cap, VIN crosses −100 mV, keeps falling ~90 ns, and bottoms at **−0.177 V** — inside the LM5164's −0.3 V by **1.70×**. The Schottky backstop barely conducts. The previous −0.068 V / 4.4× was optimistic by 2.6×: at the default step, ngspice committed the hysteretic switch on a rejected trial step and hid the dip.

Supplies are the power-up argument. The comparator runs from 5V_MAIN, which is dead whenever VIN is too low to run the buck — so it cannot hold the clamp on and short an incoming battery. The driver runs from 12V_GATE, held up ~400 ms by the boost reservoir, so the clamp keeps its gate through a whole pulse.

Clamp Schottky

100 V / 20 A class — it was 60 V; see the correction above. It carries what the TVS used to: **13.58 A peak, 14.5 mJ** on pulse 1, measured at its own ammeter in `negative_pulses.cir` rather than inferred from 150 V / 10 Ω.

A 100 V Schottky has a higher forward drop than the 60 V part `negative_pulses.cir` models (`SCHN`, BV = 60 V). That drop only matters for the comparator's propagation delay — the window before the active clamp engages — and is re-measured when the comparator is chosen.

Reverse-battery FET

P-channel, |Vds| ≥ 80 V, Rds(on) ≤ 40 mΩ at Vgs = −4.5 V, drain to the battery side, source to the load, with a **passive gate network**: 47 kΩ gate-to-ground and a 12 V zener gate-to-source.

> **CORRECTED 22 September 2026 — the sheet had drawn an N-FET, and `reverse_battery.cir` has always modelled a P-FET.** Its header says "a P-FET held on by a controller that watches the polarity", and its switch model closes on positive polarity. Found while choosing the controller the N-FET would have needed.

No controller IC checked can sit on this node. It swings from **+73.3 V** (pulse 2a, after the TVS) to **−47.8 V** (pulse 1, the bidirectional TVS's negative clamp — bidirectional precisely so a reversed battery does not forward-bias it). That is a 121 V span:

Controller	Rated	Fails by
LM74700-Q1	ANODE −65 V to **+65 V**	8.3 V on the positive side
LM5050-1-Q1	IN **−0.3 V** to +100 V	47.5 V on the negative side — and IN draws 320 µA, so a series resistor to protect it corrupts its 20 mV sense

A P-FET needs no controller: `Vgs = -V(load)`, so it is on whenever the load side is positive and off when it is pulled to zero. That **is** the block's switch model, with no IC ratings on a 121 V swing to violate.

What it gives up, recorded rather than hidden. An ideal-diode controller blocks reverse *current*; this blocks reverse *polarity*. On a dropout that *shorts* the input, bulk capacitance discharges backward through the FET. The spec carries no hold-up requirement, and the negative clamp is already sized for this FET staying on through a whole pulse — its turn-off via 47 kΩ and `Ciss` takes tens of microseconds.

The 40 mΩ ceiling gives 0.2 V at 5 A against the block's 0.3 V requirement. The model's 8 mΩ is its header's own "mid-range for a 100 V P-channel" and is optimistic; the ceiling is what `run_sim.py` checks.

Power

`PDN-BULK` — 3V3_MCU bulk capacitor

> **Chosen, 8 October 2026 (decisions log §9):** KEMET T599D157M006ATE025 (150 µF polymer tantalum, 25 mΩ max) **in series with R306, 22 mΩ 1 %.** A datasheet gives an ESR maximum and never a minimum, so the resistor is the band's floor: the pair is 21.8–72.2 mΩ across temperature and end of life. A sweep shows the target holds from about 17 to 95 mΩ, so the 50 mΩ ceiling below was conservative. `mcu_pdn.cir` checks both ends at 96 µF: 71.9 and 86.9 mΩ.

Polymer aluminium electrolytic or polymer tantalum, 150 µF, specified by an **ESR band of 20–50 mΩ across −40…+125 °C** — not by a 25 °C typical figure.

Three things about that are load-bearing:

- **The part class changes, not the value.** A wet aluminium or tantalum electrolyte loses conductivity as it freezes, roughly 4× ESR at −40 °C, which puts the PDN peak at 197 mΩ against a 100 mΩ target. A solid polymer electrolyte has no such mechanism.

- **The cold number is what gets specified.** A supplier's 25 °C ESR does not answer this block's question, and asking for it is how the original corner got in.
- **A band, not a ceiling.** Bulk ESR is also what damps the regulator-against-bulk resonance. At 20 mΩ the peak climbs back to 92 mΩ, 8 mΩ under target. "ESR ≤ 50 mΩ" would be satisfied by a 5 mΩ part that fails this block.

The 100 mΩ target is derived — 50 mV allowed ripple over a 0.5 A transient step — not a round convention. Checked by `mcu_pdn.cir`.

`SENSOR-PTC` — 5 V sensor rail, one per sensor group

> **Superseded and built, 8 October 2026 (decisions log §4, §7).** `sensor_rail.cir` checks it: VSNS peaks at 22.3 V on pulse 2a against the trackers' 45 V. The OV window is 40.52–42.08 V with 0.1 % resistors; 1 % would put the bottom at 39.8 V, inside the 40 V dump. Inrush is 0.16 A into 47 µF. Cranking headroom is 0.46 V. The controller's VS and CS pins see 4.9 V/µs through 100 Ω + 22 nF, against their 60 V/µs limit. The hold-up, C211, was chosen 8 October (Chemi-Con HXC, decisions log §9). Still open: land patterns for the TPS48110-Q1 (DGX-19) and the TPS7B4253-Q1 (TI's DDA0008J pad matches no KiCad variant). The 10 µF 50 V output capacitors are 2220, the only case KEMET makes at 50 V; they face the harness, so buy them with flexible termination.
>
> The PTCs are replaced by one **TPS7B4253-Q1** tracking regulator per group (TI SLVSCP3D, rev. Nov 2025), following 5V_MAIN to ±4 mV and limiting at 301–520 mA. The three sit behind one **TPS48110-Q1** overvoltage cut-off (TI SLUSEE5E, rev. Apr 2026, AEC-Q100 Grade 1) switching an IPD90N10S4-06, with 47 µF of hold-up after it. The cut-off opens for pulse 2a (73.3 V; the trackers are rated 45 V) and stays closed through the 40 V load dump. The text below is the PTC requirement it replaces, kept for the record.

Resettable PTC, hold current ≥ 200 mA, **R25 = 3.0 Ω nominal**.

Sized against the ceiling rather than guessed. The polyfuse has to be what clears a shorted harness, not the 5 V regulator's own current limit, so fault current must stay under 3.0 A:

```
I < 3.0 A  ⟹  Rsrc + Rptc > 5/3 = 1.667 Ω
corner = R25 × 0.8 (cold) × 0.7 (-30% spread) = 0.56 × R25
⟹  R25 > 2.78 Ω
```

At the same stacked −40 °C + −30% corner the specified part draws 2.88 A, 0.12 A under the ceiling. The larger series resistance was re-checked against the rest of the block rather than assumed harmless: healthy-

group voltage holds at 4.69 V against a 4.5 V floor, and nominal fault current stays inside the 0.3–3.0 A window the polyfuse needs to trip on.

Checked by `sensor_rail.cir`.

Drivers

`MU-ISENSE` — metering unit current sense, ECU pin 88 return

50 mΩ shunt on the low side, into **either** a current-sense amplifier feeding an ADC channel **or** a gate driver with an integrated current-regulation comparator.

The amplifier is not optional: 50 mΩ at the 0.675 A setpoint is 33.75 mV, too small to take straight to the S32K148's ADC at useful resolution.

This one is not a part-selection fix at all — it is an architecture change, and it is the only failing corner in the suite that no component choice could have answered. The coil is copper, copper moves 0.39%/°C, and over −40…+125 °C its resistance goes 7.39 → 13.82 Ω. At a fixed 50% duty the mean current follows it: **0.862 A cold against 0.462 A hot**, +27% and −32% around the setpoint. There is no better resistor to buy.

A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at −40 / 25 / +125 °C. **Closed loop only works while the regulator has duty left** — at the hot corner the coil can draw at most 13.5/13.82 = 0.977 A at 100% duty, so the setpoint sits at 69% of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop behaviour rather than reporting anything.

Checked by `metering_unit_pwm.cir`.

`EGR-DRIVER` — EGR H-bridge

A bridge driver with **internal DIR/PWM decode** — one logic input pair that cannot express both legs of a diagonal conducting at once.

Two logic lines wired straight to two diagonal switch pairs short the bridge: with both inputs asserted it draws **270 A**. Under a decoded interface the same input combination draws **4.35 A**, the motor's own stall current. The fault state stops existing rather than being avoided by firmware convention.

Current capability: headroom over a 2.6–6.4 A stall.

> **Drawn 22 September 2026 — discrete, two `AUIRS2184STR`.** No integrated bridge clears the rail: DRV8873-Q1, the candidate, has a 40 V VM absolute maximum against 73.3 V, and the class caps at 40–45 V. So four **100 V** N-FETs (≤ 20 mΩ, headroom over the 6.4 A stall), and the decode this tag requires comes from the **driver's structure** rather than a logic

> gate: AUIRS2184S has **one input per leg** — IN high puts HO on, IN low puts LO on — with cross-conduction prevention and ~400 ns deadtime. Each leg's high and low are exclusive by construction, so the 270 A state, both inputs high, becomes a **high-side brake**.
>
> | EGR_EN | IN1 IN2 | bridge |
> |---|---|---|
> | high | 1 0 / 0 1 | forward / reverse |
> | high | 1 1 / 0 0 | high-side / low-side brake |
> | **low** | any | **COAST** |
>
> **Coast is asserted by copper.** Both drivers' SD̄ share `EGR_SD`, pulled down 10 kΩ, so with `PTB10` Hi-Z at reset both legs are off and the return spring closes the valve. A kill FET on the same node gives coast on `GATE_KILL` without the MCU. One 20 mΩ shunt in the shared low-side return feeds an INA181A1-Q1: 2.56 V at a 6.4 A stall.
>
> Pinout and logic are from the IR2184 datasheet (same die) and the AUIRS2181(4)S family table; **the automotive part's own datasheet was not retrieved** — confirm against it at sourcing.

Checked by `egr_hbridge.cir`, which keeps the naive wiring as pinned evidence of why the requirement exists.

`GATE-PULLDOWN` — fail-safe gate pulldowns, six driver gates

470 Ω, gate to source, on every driver gate.

The largest standard value that still holds a gate below Vgs(th) = 1.0 V against Miller coupling at Crss = 500 pF. The commonly-reached-for 10 kΩ fails even at the *low* end of the memo's own Crss envelope — 1.155 V at 50 pF already exceeds the threshold.

Cost, stated because it is real: 21.3 mA per gate held at 10 V, 1.28 W across all six.

Checked by `supervisor.cir`.

Signal

`CAN-TERM` — CAN split termination, Rah / Ral

60.0 Ω **thin-film**, initial tolerance **±0.1%** or better, tempco **±50 ppm/°C** or better.

Ordinary standard-grade thick-film runs 100–200 ppm/°C. At 200 ppm/°C over the full −40/+125 °C span, both resistors shifting together, the differential impedance reaches 123.95 Ω against J1939's own 120 Ω — and 120 Ω here is what the standard specifies, not a round convention, so missing it is a real finding.

The contrast is a part class, with no meaningful cost or availability penalty at this quantity. With both mechanisms stacked the adverse way (60 × 1.00825 × 1.001 = 60.5555 Ω) the specified part lands at 121.10 Ω, inside the ±2 Ω window with 0.90 Ω to spare. Checked at the **worst case**, not the nominal — so passing means the requirement is sufficient, not merely that it helps.

Checked by `can_termination.cir`.

`VR-CLAMP` — crank input clamp diodes, four off

Reverse leakage **≤1 µA at 125 °C**. Forward drop is explicitly *not* a requirement.

This inverts the rule every earlier clamp on this board was chosen by. The negative-clamp Schottky's 0.547 V at 15 A decomposes into a junction term and a bulk term, and that decomposition is what ruled out paralleling junction devices for the power path. These four diodes conduct only on a fault, into a 4.7 kΩ series resistor that is already doing the limiting — what matters is what they do when they are **off**.

A reverse-biased diode's leakage flows out of the 4.489 kΩ Thévenin at the comparator input and appears there as an offset voltage, against a hysteresis window of ±198 mV:

Class	Leakage at 125 °C	Offset	Against the window
BAT54S-class Schottky	~200 µA (≈2 µA at 25 °C, ~100× hotter)	0.90 V	**4.5× the whole window** — latches
BAV199-class low-leakage silicon	<1 µA (5 nA max at 25 °C)	<4.5 mV	2% of the window

The two diodes on a leg are reverse-biased symmetrically by `VR_BIAS`, and the legs are matched, so leakages partly cancel and what survives is partly common-mode. That is not a thing to defend a speed input on, so the requirement is stated instead.

`vr_conditioner.cir` does **not** model this — its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Found while drawing `hw/speed_inputs.kicad_sch`.

The same tag covers the **differential TVS across the pair**: bidirectional, standoff ≥48 V. The standoff has to clear the *signal*, not just the rail — VR amplitude is proportional to speed, about 20 V peak at 1500 rpm, so a 24 V part would clip a runaway engine's own crank signal at the one moment the ECU most needs to keep counting teeth. 48 V clears 3600 rpm.

`VR-COMP` — crank zero-cross comparator

Single **3.3 V** supply, **push-pull** rail-to-rail output, propagation delay **≤1 µs**. Input common-mode range: **mid-rail only**.

> **Chosen 22 September 2026:** `TLV3201AQDCKRQ1` — the same part as the negative clamp's comparator, so one part number serves both. 2.7–5.5 V, push-pull, input range (VEE) − 0.2 V to (VCC) + 0.2 V, **50 ns max** against the 1 µs requirement, 5 mV offset against a ±198 mV window.

Three of those four requirements are unusually easy, deliberately:

- The output swing *is* the hysteresis window (3.3 V × 4489/(4489+33000) = 0.395 V), so an open-drain part would make the window depend on its pull-up value and its V_{OL}.
- Biasing the floating VR coil to `VR_BIAS` means the inputs never leave mid-rail, so **no common-mode range including ground is needed** — the usual hard constraint on a zero-cross detector, avoided by biasing the sensor rather than level-shifting it.
- 1 µs is 0.009 crank degrees at 1500 rpm on a 60-tooth wheel.
- Input offset voltage and its tempco are non-issues: 10 mV is 5% of the window.

Checked by `vr_conditioner.cir`, which establishes the topology (a fixed 5 V threshold produces edges at 1500 rpm and **none** at cranking speed) and the ±0.2 V window the E24 network above lands on.

`BOOST-BLEED` — injector boost reservoir bleed

22 kΩ, 1 W, across the reservoir. Since 8 October 2026 the reservoir is two 82 µF / 80 V cans in series (41 µF, decisions log §9), and the bleed is **2 × 11 kΩ 1 W**, one across each can (R401, R435). Same total current and power; the split also balances the stack, at 52.4 V per can worst case.

The charger regulates from **one side only**: `boost_converter.cir`'s `Bchg` adds 50 mA while the rail is below setpoint and adds nothing above it. Nothing else in the circuit can remove charge, so any excursion above 100 V is permanent. `injector_turnoff.cir` measured the mechanism that produces one — a hold-current cutoff, whose energy came from the **battery** through the high side's diode-OR and so has no boost-side draw to net against — and its RESULT NOTE asked for this resistor by name.

Sizing is not a duty-cycle argument; it is "one event is gone before the next cylinder fires":

event charge	10 A decaying to 0 over 20 µs = 100 µC
on 41 µF	2.45 V of rail rise
above setpoint	charger is **off**, so the whole 100 V / 22 kΩ = 4.55 mA discharges it
time to clear	**21.4 ms**, against **26.67 ms** between cylinders at 1500 rpm
cost	**455 mW**, burned continuously whenever the rail is up

Checked by `boost_converter.cir` arrangement C, which runs two identical reservoirs — one bare, one bled — through three such events at the real cylinder spacing. Bare, the rail stacks to 107.4 V and stays there.

A comparator-gated shunt costs nothing at idle and is the alternative. It is more parts, and it is a part nobody has chosen.

Over a complete injection event there is no pumping at all — the peak phase draws 342 µC from the reservoir and end-of-injection recirculation returns roughly 100 µC of it. This resistor is not there for normal running. It is there because one-sided regulation has no answer for an excursion from *any* cause.

`INJ-RECIRC` — injector recirculation diodes, five off

150 V, 20 A, ultrafast. Three on the low sides (one per cylinder, into the boost reservoir) and two on the high sides (the battery diode-OR).

Research memo 10 names the class. Not a Zener active clamp across the FET — Nexperia AN50003 rates repetitive active clamp reliability-"Questionable" for gate-oxide wear-out. Not a bare freewheel to battery, which is too slow: the same problem the boost rail exists to fix on the opening side.

Sized by the pulse, not the average. `injector_turnoff.cir` measures 28.7 W instantaneous during turn-off against roughly 3.6 W averaged over all nine events per second — a factor of eight. The average would never have sized this part.

Chosen parts

These are the entries that have moved from "a class" to "a part number", each against a datasheet that was actually retrieved and read. They are listed separately from the requirements above because they are answers, not constraints.

Function	Part	Why this one
Buck pre-regulator	**LM5164DDAR**	100 V input, 1 A, 6–100 V range. See below — it replaces a 60 V part that can no longer survive this input.
3V3 regulator	**TLV76733QWDRBRQ1**	1% over load *and* temperature. The supervisor makes that load-bearing.
Supervisor	**TPS3850G33DRCT**	±4% window variant, so its undervoltage trip clears the MCU's own LVD by 143 mV rather than 44 mV.
Injector boost controller	**TPS40210QDGQRQ1**	Chosen on its **200 ns maximum off-time** — the one spec that makes 94% duty reachable at the 6 V cranking corner. See below.

Function	Part	Why this one
Gate drivers, all eight gates	**AUIRS2181STR** ×5	**No cross-conduction interlock** — the injector's high and low switches are in series with the coil and must both be on. See below.
Difference amplifiers	**INA592IDR** ×5	Native G = ½, 88 dB CMRR min against a 60 dB floor the discrete network could not reach. **Not AEC-Q100** — see below.
Barometric sensor	**MPXHZ6115A6T1**	Spec §7 deviation 2, on-PCB behind a vented port. AEC-Q100 **not stated** in its datasheet — see below.
EGR bridge drivers	**AUIRS2184STR** ×2	One input per leg: each leg's high and low exclusive by construction. See `EGR-DRIVER`.
Negative-clamp control	**TLV3201AQDCKRQ1** + **UCC27517AQDBVRQ1**	A low-side ideal diode: engage below −100 mV, release above −6.8 mV. See `NEG-CLAMP`.
Current-sense amplifiers	**INA181A1QDBVRQ1** ×3	VCM reaches −0.2 V, so it sits across a ground-referenced shunt; supplied from 3V3_MCU so its output is ADC-safe by construction.
Kill FETs and reset inverter	**2N7002BK** ×10	60 V, ±20 V Vgs, Vth 1.1–2.1 V, 2 Ω max at Vgs 5 V against the 5 Ω `supervisor.cir` assumes. AEC-Q101. Needs `GATE-KILL-CLAMP`.
GATE_KILL clamp	**BZX84-C12-Q**	11.4–12.7 V at 5 mA, +10 mV/K max. AEC-Q101. The plain BZX84 series is no longer automotive-qualified; the `-Q` suffix is.
Relay FETs	**PMV280ENEA** ×2	100 V, ±20 V Vgs, 432 mΩ max at Vgs 4.5 V. AEC-Q101. Driven through a buffer — see `RELAY-GATE`.
Relay gate buffers	**CAHCT1G125QDBVRQ1** ×2	SN74AHCT1G125-Q1 on 5V_MAIN: VIH 2 V reads a 3.3 V GPIO, VOH 4.4 V min drives the FET. AEC-Q100.
Reverse-battery FET	**SQD50P08-25L**	−80 V P-channel, ±20 V Vgs, 31 mΩ max at −4.5 V against a 40 mΩ ceiling. AEC-Q101, TO-252.
Pin 21 filter inductor L101	**7447709220**	Würth WE-PD 1210, 22 µH ±20%, IR 5.3 A, Isat 6.5 A typ, RDC 28 mΩ max, SRF 10 MHz typ, 120 V, AEC-Q200 Grade 1, −40 to +125 °C. Datasheet rev 003.001 (2024-05-08).
Pin 21 second stage L102	**744314047**	Würth WE-HCI 7050, 0.47 µH ±20%, IR 18 A, Isat 9 A typ, RDC 1.35 mΩ, SRF 235 MHz typ, AEC-Q200 Grade 1, −40 to +150 °C. Datasheet rev 004.001 (2025-02-06).
Pin 21 fuse F101	**0437005.WRA**	Littelfuse 437A, 1206 fast-acting, 5 A, 63 V DC, 50 A interrupting, melting I^2t 2.30 A^2s, 15.9 mΩ, AEC-Q200. Datasheet revised 09/08/21. 79 % re-rating at 105 °C read off its curve.
Sensor trackers, ×3	**TPS7B4253QDDARQ1**	TI TPS7B4253-Q1, SLVSCP3D (rev. Nov 2025): tracks ADJ to ±4 mV, 300 mA, limit 301–520 mA, dropout 520 mV max at 200 mA, IN −40…45 V, OUT −1…45 V. AEC-Q100. DDA PowerPAD-8; land pattern to draw.
Sensor-supply OV cut-off	**TPS48110AQDGXRQ1**	TI TPS4811-Q1, SLUSEE5E (rev. Apr 2026): 100 V abs max, OV 1.16–1.20 V rising, 2.6–4 µs to gate off, 4 A pull-down, 60 V/µs limit on drain-side pins. AEC-Q100 Grade 1. DGX VSSOP-19; land pattern to draw. Chosen over LTC4368 (not AEC-Q100): decisions log §7.
Boost inductor L401	**7447709151**	Würth WE-PD 1210, 150 µH ±20%, IR 2.1 A, Isat 2.7 A (10 %) / 3.3 A (30 %) typ, RDC 200 mΩ max, 120 V, AEC-Q200 Grade 1. Datasheet rev 003.001 (2024-05-08). Decisions log §8.

Function	Part	Why this one
Boost VDD reference	**BZG03C39-HM3**	Vishay BZG03C-M, 37–41 V at 10 mA, ZZ 40 Ω, +0.06–0.11 %/K, AEC-Q101, SMA. Doc 86574 Rev. 1.0.
Reverse-battery FETs, pins 04/06	**SQM120P10-10m1L** ×2	−100 V P-channel, ±20 V Vgs, 10.1 mΩ max at −10 V, 20.5 mΩ at 175 °C, 15.0 mΩ at −4.5 V, RthJA 40 °C/W. AEC-Q101, TO-263. Vishay doc 76943 Rev. A, retrieved 8 Oct 2026. `POWER-PATH`.
100 V N-channel	**IPD90N10S4-06** ×8	Negative clamp, metering, four EGR, two injector battery switches (60 V class needed; one part number instead of two). 6.7 mΩ max at 10 V, ±20 V Vgs, AEC-Q101, TO-252. The logic-level sibling `-S4L-06` was rejected for its ±16 V Vgs: every one of these gates is driven from 12V_GATE.
Injector 150 V FETs	**SQM85N15-19** ×5	Two boost-side high sides, three low sides. 19 mΩ max at 10 V — the model's 20 mΩ. ±20 V Vgs, AEC-Q101, TO-263. Qg 120 nC max sets the bootstrap droop: 126 mV.
Boost switch	**DMN15H310SK3**	150 V, Vth 3 V max, 350 mΩ max at Vgs **4 V** — specified below the 5.7 V it gets while cranking. Qg 8.7 nC typ. AEC-Q101, TO-252. Cranking duty with its drop, hot: 94.8% against 97.0% available.
Injector recirculation, blocking, freewheel	**VS-16EDH02HM3** ×7	200 V, 16 A avg, 250 A surge, trr 32 ns, AEC-Q101. SMPD (TO-263AC): **no KiCad 7 footprint — drawn in phase 3.** Average current per diode is tens of mA; the rating is for the 18 A peaks.
Boost diode	**VS-2EMH02HM3**	200 V, 2 A, trr 25 ns, AEC-Q101, SMA.
Bootstrap and relay flyback diodes	**VS-1EMH02HM3** ×8	200 V, 1 A, 50 A surge, AEC-Q101, SMA. The relay flybacks needed only 100 V; one part number instead of two.
Metering freewheel	**VS-3ECH02HM3**	200 V, 3 A, 130 A surge, AEC-Q101, SMC.
Negative-clamp backstop Schottky	**SS20PH102HM3**	100 V, 20 A, 310 A surge, AEC-Q101, TO-277A. Stock footprint puts the cathode on pad 3; the project's `TO-277A_K1_A2` renumbers it (`hw/gen_footprints.py`).
Battery TVS	**SMCJ43CA-HM3**	43 V standoff, 47.8–52.8 V breakdown, 69.4 V clamping at 21.6 A, 1500 W, AEC-Q101.
Crank-pair TVS	**SMAJ48CA-HE3**	48 V standoff, 53.3–58.9 V breakdown, 400 W, AEC-Q101.
12 V clamps	**BZX84-C12-Q** ×2, **BZX84-C13-Q**	GATE_KILL, reverse-FET gate, 12V_GATE reference. SOT-23 with pin 3 as the cathode, so the project's `SOT-23_Zener_K1_A2` footprint is used. The GATE_KILL zener's first footprint, the stock SOT-23, had its anode where the symbol puts the cathode.
VDD clamp	**BZG03C36-HM3**	See `BOOST-VDD-CLAMP`: it replaced a 43 V clamp that did not clamp.
Buck inductor	**Würth 7447714330**	WE-PD 1050, 33 µH, Isat 2.9 A typ against a 1.18 A worst peak (1.72× after a −30% hot derate), 79 mΩ, AEC-Q200 Grade 1. This closes the Isat gap `buck_preregulator.cir` left open. Land pattern not in KiCad 7.
12V_GATE pass / limit	**PZTA42-Q**, **PMBT3904-Q**	See `GATE-RAIL`. PMBT3904-Q is SOT-23 B-E-C, so it uses the `Q_NPN_BEC` symbol.
3V3_MCU bulk (C313)	**T599D157M006ATE025** + 22 mΩ 1 % (R306)	KEMET polymer tantalum, 150 µF, 6.3 V (5.4 V at 125 °C), 25 mΩ max, −55/+150 °C, AEC-Q200, 7343-31. The resistor is `PDN-BULK` 's ESR floor; see decisions log §9.

Function	Part	Why this one
Sensor hold-up (C211)	**HHXC630ARA470MHA0G**	Chemi-Con HXC hybrid polymer, 47 µF 63 V, 8 × 10 mm SMD, 40 mΩ max, Z(−55)/Z(20) ≤ 2 at 100 kHz, 4000 h at 125 °C, AEC-Q200. 1.58× over the 40 V load dump; 26.3 µF tolerance-and-aged corner in `sensor_rail.cir`.
Boost reservoir (C401, C420)	**HHXC800ARA820MJC5G** ×2, in series	Chemi-Con HXC hybrid polymer, 82 µF 80 V, 10 × 12.5 mm SMD, 28 mΩ max, cold ratio ≤ 2, 2.1 A rms at 125 °C, AEC-Q200. 41 µF / 160 V stack, balanced by the 2 × 11 kΩ bleed. Chosen over one 160 V wet can because no part above 100 V publishes a cold ESR (decisions log §9).

`PASSIVE-PACKAGES` — 188 resistors and capacitors, sized by what is across them

`hw/assign_passives.py` gives every chip resistor and capacitor its package, and `hw/check_netlist.py` fails if a part is added or re-valued without re-running it.

The voltage comes first. Each net carries a continuous and a peak voltage: battery nets are 24 V jump-start and 73.3 V pulse 2a, the boost rail is 104 V. The voltage across a part is the worse of its two nets, with two exceptions. Inside a bootstrap domain, a part sees only the 13.6 V gate swing. And `PAIR_V` overrides the few series strings and AC-only parts where "the worse net" over-states what is across them. Both are keyed by net names, which is why several internal nodes were given labels: the auto-names `Net-(R407-Pad2)` move whenever a part is added ahead of them.

- **Resistors: Vishay D/CRCW e3 ratings** (AEC-Q200, standard mode). The peak voltage must clear Umax by the rail table's 1.09×. The continuous V^2/R must fit the power rating derated to 0.588× at a 105 °C board. Nothing is smaller than 0603.
- **Capacitors: KEMET C1023 X7R AUTO** (AEC-Q200). The rating must clear the peak voltage and 1.5× the continuous one, and nothing up to 1 µF is rated below 25 V. The part picks the smallest case that KEMET's waterfall table actually lists at that rating. The table is parsed by x-coordinate into `hw/lib/kemet_x7r_auto.json` (`hw/tools/parse_kemet_waterfall.py`), because the PDF's text layout compresses the data columns and misreads them.

Result: 121 × 0603, 36 × 0805, 5 × 1206, 7 × 1210 and 12 × 2010, plus the stated 2512 parts and the shunts.

Things this pass found:

- **4.7 µF at 100 V is not in the automotive X7R range**; 1210 stops at 50 V. Three places needed it: the EMI filter's C1, the buck input and the boost input. Each is now two 2.2 µF, 100 V, 1210 parts. `emi_filter.cir` models the pair as 4.4 µF with ESR and ESL halved. Its cold corner is 0.6 dB worse, and the check was re-pinned.

- **The twelve 470 Ω gate pulldowns land in 2010.** At 100% duty (metering, EGR) 12 V across 470 Ω is 0.31 W per gate, as well as 25 mA of driver current. They were sized for Crss = 500 pF, and the worst chosen FET is 300 pF, so 1 kΩ may now serve. That would need a re-run of `supervisor.cir` claim 6 before it could be claimed.
- **DC bias on bulk MLCCs is not modelled.** Two parts are affected: the 47 µF 10 V buck output and the 10 µF 6.3 V on 3V3_MCU. Both lose a large part of their capacitance at their working voltage. The 1.5× rule does not capture that.

`POWER-PATH` — what the input filter and reverse FET actually carry (CLOSED 8 October 2026)

Built. Pin 04 now feeds `VBAT_INJ` and pin 06 feeds `VBAT_EGR`. Each feed runs through its own SMCJ43CA TVS, an **SQM120P10-10m1L** reverse P-FET (passive gate, as Q1), an SS20PH102HM3 Schottky backstop and 2 × 2.2 µF 100 V. The reasons for the split are in `docs/decisions-log.md` §2a. `power_path.cir` checks these figures:

	EGR, pin 06	Injector, pin 04
Load	6.4 A stall, continuous	10 A hold 3 ms / 26.67 ms + 0.98 A boost
FET drop at 175 °C RDS(on)	0.131 V	0.225 V peak
FET dissipation	0.84 W	0.30 W (3.8 A RMS)
Tj at a 105 °C board	139 °C	117 °C
Tj while cranking (Vgs −6 V)	155 °C (RDS(on) inferred)	—

The netlist gate checks that each rail leaves its own FET, reaches its own loads, and that VBAT_PROT no longer reaches either power stage.

What this leaves open:

- **L101, L102 and F101 are chosen (8 October 2026)** for what pin 21 now carries, stated as an upper bound of **2.95 A**. The bound is 1.75 A for the buck, from the LM5164's IPEAK1 maximum, since its average input current cannot exceed its peak limit. Then 0.30 A for two relay coils at 24 V, 0.86 A for the metering freewheel, and 0.04 A for the boost VDD feed. `run_sim.py` checks all of it under `emi_filter`, `transient_clamp` and `negative_pulses`:

Part	Choice	Against the 2.95 A bound
L101	Würth **7447709220**, 22 µH WE-PD	Isat 6.5 A typ (2.2×); 117 °C on a 105 °C board, under its 125 °C maximum
L102	Würth **744314047**, 0.47 µH WE-HCI 7050	Isat 9 A typ (3.05×)
F101	Littelfuse **0437005.WRA**, 5 A 437A	3.16 A continuous at 105 °C (1.07×); pulse 1 I^2t 16 % of melting, pulse 2a 2.5 %

Thin margins, stated:

- **F101's 1.07×** is against a bound the buck cannot actually sustain; the estimated real load is 0.25–0.5 A at the input. 5 A is also the largest 437A rating still rated for 63 V, since 7 A is 35 V, under the 40 V load dump. If the fuse opens on overload, the board loses power, which is the fail-safe direction.
- **The 20 % pulse ceiling** for the repeated pulse 1 is a fuse-fatigue rule of thumb, not a retrieved figure. Littelfuse pulse-cycling data is open.
- **The EMI filter's margins at 150 kHz and 400 kHz fell** to 0.56 dB and 0.73 dB once the model carried the real inductors' RDC and self-resonance (they were 1.17 and 1.55 dB with ideal inductors). Lf is 0.47 µH, not the 0.5 µH it was modelled at. The gates are still the unvalidated placeholders `emi_filter` already flags.
- **Conducted emissions from the EGR PWM and the injector chopping** no longer pass through the pin-21 filter. They were never modelled through it either, but the new rails have only 4.4 µF of ceramic each. A CISPR 25 check of these two rails is open.
- Anything placed on VBAT_INJ or VBAT_EGR has to tolerate about −0.5 V on pulse 1. The Schottky holds the rail there, and there is no active clamp on these rails. The present loads are switch and bridge FETs, a diode and an inductor, and all of them do.

The original requirement follows.

Every high-current load on the board draws from VBAT_PROT: the injector battery switches (10 A hold), the EGR bridge (up to 6.4 A stall) and the relay coils. VBAT_PROT is fed through connector pin 21, then the 20 A fuse, the 22 µH + 0.5 µH EMI filter, the TVS and the reverse FET. Two blocks sized parts on that path, and neither was written with this load in mind:

- `reverse_battery.cir` states its claim at **5 A** (0.2 V).
- `emi_filter.cir` was written for the buck's 400 kHz noise and never states a DC current.

A 22 µH inductor that does not saturate at 15–20 A is a large part.

The connector has two more battery feeds, pins **04** and **06** (`V_BAT_1R/2R` , each fused 10 A in the loom). The spec says they "feed high-current loads [?]", and the board uses them only as ADC senses. Moving the injector and EGR supply onto 04/06, with its own protection, would leave pin 21 feeding the electronics: a few amps through the filter. Keeping everything on pin 21 means resizing the filter, the fuse and the reverse FET for the full load, and relying on one contact to carry it.

This is an architecture decision. L101, L102 and F101 stay unchosen until it is made.

`BOOST-SLOPE` — current-mode stability constrains the boost inductor (CLOSED 8 October 2026)

Built (decisions log §2b, §8). The TPS40210-Q1 runs from its own output through a follower: 34.3–45.0 V, with a 39 V zener, PZTA42-Q, 2 × 3.3 kΩ and a PMBT3904-Q limiter, as 12V_GATE. The battery feeds it only at start-up, through 180 Ω, the 36 V clamp and a Schottky. **L401 is Würth 7447709151, 150 µH**, and the sense resistor is **62 mΩ**. `run_sim boost_converter` checks, at the worst corner (VDD 34.3 V, 120 µH, oscillator −20 %, 6 V in):

Check	Result
Equation 9, TI's 80 % ceiling	69.4 mΩ: 62 mΩ inside by 1.12×
Equation 49, no false trip	65.1 mΩ: inside by 1.05×
Current limit	1.94–2.90 A: 7 % past L401's 10 % saturation, 14 % under its 30 % point, in a fault only
Start-up VDD at 6 V	4.77 V against the 4.5 V threshold
Standing load on the boost	17.6 mA; peak-phase recovery 10.6 ms against 26.7 ms

Stated rather than hidden: at twice the switch FET's typical gate charge (no maximum is given), start-up margin is 17 mV. Before the output passes about 36 V, the ramp is still the battery's.

The original finding follows.

TPS40210-Q1 applies a fixed internal ramp of fSW × VDD / 20. TI's Equation 9 (SLVS861F §7.3.3) therefore caps the sense resistor at `VDD·L·fSW / (60·(VOUT + VD - VIN))`, and TI recommends 80% of that.

	330 µH, VDD ≈ battery
13.5 V in	ceiling 99.7 mΩ (at 80%): 82 mΩ **passes**
6 V cranking, VDD 5.16 V	ceiling 44.9 mΩ (at 100%): 82 mΩ **fails**, 1.83× over

The failure is subharmonic oscillation while cranking. TI says the voltage loop still regulates, but ripple grows. The inductor makes it worse. A 330 µH part rated to saturate above the 2.2 A current limit is not a part that exists in SMD: the search found 330 µH parts at 0.2–0.4 A. The buildable direction is a smaller inductor, which lowers the ceiling further: 100 µH allows 30 mΩ.

TI's own remedy is to feed VDD from the output side for a steeper ramp. VDD's 52 V absolute maximum rules out connecting it to the 100 V rail directly; it would need a clamped feed, such as the 36 V zener that is already there. Either that, or accept subharmonic operation while cranking. **L401 stays unchosen** until one of those is decided. The candidate at 100 µH is Würth **7447709101**: WE-PD 1260, Isat 3.1 A typ, 110 mΩ max, AEC-Q200 Grade 1.

`SENSOR-PTC` — real PTCs are not specified the way this was written (OPEN)

The requirement below asks for R25 = 3.0 Ω nominal with a −30% spread. PTC datasheets do not work that way. Bourns MF-USMF020 (AEC-Q200) is specified as a **range, 0.40–5.00 Ω**. At its 0.40 Ω minimum it contributes 0.14× of the 2.78 Ω the fault-current ceiling needs. Its hold current also derates from 0.20 A at 23 °C to **0.10 A at 85 °C**, half the requirement.

The fault-current ceiling has to come from a fixed series resistor, or from the regulator's own limit, with the PTC only doing the tripping. Its hold current then has to be chosen at the hot corner.

`LOWV-CLAMP` — the 3.0 V and 3.3 V input clamps

Analog and cam inputs: closed 22 September 2026. The ten 3.3 V zeners became **BAV199-Q**. That part is a low-leakage double diode, 80 nA max at 150 °C, AEC-Q101. Its common pin sits on the signal, K2 goes to 3V3_MCU and A1 to ground. It is drawn as one 3-pin part, so a pair is one package; the speed input's split halves were redrawn the same way.

These inputs clamp only on a fault: a channel's normal maximum is 3.08 V, below the rail. A rail clamp in that position brings a problem the zener had hidden. The BAV199-Q's upper diode has a high Vf, which is the price of low leakage: 0.9 V at 1 mA. On a 40 V harness short, the MCU pad's own ESD diode sits in parallel with it and would carry most of the 3.6 mA fault current. That is over the S32K148's ±3 mA injection limit. So every clamped channel now has **1 kΩ between the clamp node and the pad**. The fault node rides at 4.33 V, the pad at 3.93 V, and **0.40 mA** is injected. That is checked in `sensor_ratiometric` and `cam_frontend`, which now model the diode pair, the 1 kΩ resistor and the pad's ESD diode.

The same models exposed what the zener had been doing in normal operation. At a 5.25 V rail the cam input sits at 3.23 V, and the 3.3 V zener model was loading it by **84 mV**. The old check had pinned that loading as the expected value.

Discrete inputs: closed 22 September 2026, by removing the clamp. The five switch inputs are now NPN buffers.

- **Circuit:** 10 kΩ input, 4.7 kΩ and 4.7 µF from base to ground, a BAV199-Q lower diode on the base, and a PMBT3904-Q whose collector is the MCU pad. The pad has a 10 kΩ pull-up to 3V3_MCU.
- **Why:** a 6–40 V input cannot be divided into logic range without clamping continuously. That rules out a zener (soft knee), a rail clamp (the pad parks at 3.9 V) and a shunt reference (its ~35 µA bias, TI SNOS633N Fig. 5-4). The buffer needs no clamp: the pad sees only 0–3.3 V.

- **Checked at the worst corner**, per the owner's rule that safety and reliability come first:
- hFE 30 (the datasheet minimum of 60, halved for −40 °C) still pulls the pad to 0.03 V from a 6 V input.
- A typical part at 125 °C does not turn on below a 1.3 V input.
- On −24 V reverse battery the base is held at −1.05 V, against a 6 V VEBO.
- One pad edge for a 5-chatter bounce.
- **Wetting current:** 1.1 mA through a closed contact at 12 V. The 100 kΩ first drawn gave 0.11 mA, at which non-gold contacts oxidise open in the field.
- **Logic is inverted**: pad LOW = input active, or contact open on the two contact-to-ground channels. `docs/pinmap.md` §1 records it.
- **Failure modes are for the firmware spec.** A transistor that fails open leaves the pad stuck high, and a C-E short leaves it stuck low. One polarity cannot make every channel fail safe.

Residual: the trip loop keeps its clamped analog front end, because its healthy band *is* the 3.0 V clamp level. It is also still blocked on a connector pin.

`GATE-KILL-CLAMP` — 12 V zener on GATE_KILL

`GATE_KILL` is pulled up by 47 kΩ to **VBAT_PROT**, not 3V3_MCU, and for a good reason: a pullup on the rail being supervised has no supply during the brownout it exists to catch. But nothing on the net draws DC: it is nine FET gates and a reset inverter's drain. So with the inverter off, the net follows the battery rail, and that rail reaches **73.3 V** on pulse 2a. That is 3.7× the ±20 V Vgs rating of any small-signal FET.

This was not caught while the kill path was drawn. It surfaced on 22 Sep 2026, when the first kill FET was being chosen and its Vgs rating had to be checked against what drives its gate.

A BZX84-C12-Q from GATE_KILL to ground holds the net at 12.7 V max, or 13.7 V at 125 °C. Its current at the pulse peak is 1.3 mA, and the 47k dissipates 78 mW for the pulse's duration. Below 12 V, which includes cranking, the zener is out of circuit and GATE_KILL is the battery, which is what the pullup is there for. The numbers are checked in `run_sim` `transient_clamp`.

`RELAY-GATE` — the relay FETs' gates at 5 V, through a buffer

The relay sheet drove each FET's gate straight from a 3.3 V GPIO, and its field note said "fully enhanced at 3.0 V". No part backed that claim. The 100 V automotive small-signal FETs checked were PMV280ENEA, DMN10H220LQ and BSS123N. Every one of them specifies Rds(on) at Vgs = 4.5 V and no lower, and their Vth runs up to 2.5–2.8 V. That leaves 0.5–

0.8 V of overdrive, less when cold, and the datasheets guarantee nothing there. The 100 V class is not optional here, because the drain is on VBAT_PROT through the coil.

So an SN74AHCT1G125-Q1 on 5V_MAIN sits between the pin and the gate. Its TTL inputs read 3.3 V logic (VIH 2 V at VCC 4.5–5.5 V), and its output is 4.4 V minimum. OE is tied low. A 100 k pulldown on the input keeps the relay off while the GPIO is Hi-Z from reset. The existing 10 k pulldown on the gate covers the case where 5V_MAIN is absent. At 84 mA of coil current, the PMV280ENEA drops 36 mV.

`INA592IDR` — the difference amplifiers, and a stronger argument

TI SBOS914F, October 2018, revised April 2021. Retrieved and text-extracted. Five, one per differential channel on `sensors_analog`.

Gain	native **G = ½** (±IN inputs, SENSE tied to OUT)
CMRR	**88 dB minimum**
Gain error	±0.03 % max
Supply	4.5–36 V single; runs from 5V_MAIN
Output	within 220 mV of each rail
Input	common mode to 3·(V+) − 2·REF = 15 V at G = ½; only absolute maximum is 10 mA, and the ~12 kΩ internal network holds a 40 V harness short to about 3.3 mA

Why a difference amplifier and not an op-amp. The requirement was ≥ 60 dB CMRR, the floor `sensor_differential.cir` tried. The drawn design — an op-amp with a discrete 26k/16k network — could not reach it: 0.1 % discrete resistors give about 48 dB. The network has to be matched on-die. G = ½ also puts a 0.5–4.5 V sensor at 0.25–2.25 V, straight into the ADC.

And the argument got stronger with a real number. The block had modelled 80 dB, marked INFERRED, at gain 16/26. With the chosen part's **minimum** 88 dB at G = ½ — and both paths' errors referred to the sensor, since they no longer share a gain — differential now beats single-ended by **394×** on a 1 Ω corroded return, up from 90×.

A **1 kΩ** now sits between each amplifier and its ADC tail: the amplifier runs from 5 V and can rail at 4.78 V on a harness fault, where the 3.3 V zener at the pin would otherwise take the amplifier's full short-circuit current. With 1 kΩ it takes 1.5 mA.

Not AEC-Q100, and nothing better was found. INA2132 was the dual alternative, but its "G = ½" (datasheet Figure 12) is a single-ended attenuator; its matched network only gives a *difference* G = 1, which pushes a 4.5 V sensor past its 4 V swing on 5 V. No qualified difference amplifier with a differential G = ½ turned up.

`MPXHZ6115A6T1` — the onboard barometric sensor

NXP (Freescale) MPXA6115A series data sheet, Rev. 7.3, April 2015, retrieved and text-extracted. Spec §7 deviation 2 adds it on purpose: "on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction."

Range	15–115 kPa absolute
Transfer	`VOUT = VS × (0.009·P - 0.095)` , P in kPa
Supply	4.75–5.25 V, 10 mA max
Accuracy	±1.5 % of span, 0–85 °C; compensated −40…125 °C
Package	super small outline, case 98ARH99066A — a footprint KiCad ships

Supplied from **5V_MAIN**, which `SENSOR_5V_MON` already measures — so the reading is ratiometric-correctable with no new channel. Through the same 10k/16k divider every 5 V channel on the sheet uses: **2.51 V at sea level** (101.3 kPa), 1.92 V at 2000 m. No zener at the MCU pin: every other channel on that sheet arrives from the harness; this never leaves the board.

AEC-Q100 is not stated in the retrieved datasheet. Infineon's KP236 is listed automotive-qualified with a 40–115 kPa range and would be the choice, but its datasheet did not resolve by any route this session — Mouser timed out twice, Infineon's own host redirected. It is a **footprint change**, not a drop-in, so the choice is recorded rather than guessed: confirm qualification at sourcing, or retrieve the KP236 sheet and swap.

The **vented port** is a mechanical requirement the schematic cannot carry.

`INA181A1QDBVRQ1` — one part, one gain, three channels

TI SLYS018F, April 2018, revised October 2024. Retrieved and text-extracted.

Common mode	**−0.2 V to 26 V** — covers a ground-referenced shunt
Gain	20 V/V (A1), ±1 % max over temperature
Offset	**±150 µV max at VCM = 0 V**, 1 µV/°C max drift
Bandwidth	350 kHz (A1)
Supply	2.7–5.5 V; output swings to VS − 0.02 V
CMRR	84 dB min

Channel	Shunt	Full scale	Where it sits
Injector bank A → `PTC15`	5 mΩ	1.80 V at 18 A peak, 1.00 V at 10 A hold	ADC0
Injector bank B → `PTD19`	5 mΩ	same	ADC1 — simultaneous with bank A

Channel	Shunt	Full scale	Where it sits
Metering → `PTC16`	50 mΩ	675 mV at the 0.675 A setpoint, 844 counts	ADC0

The gain is chosen for the injector. 1.80 V at the peak leaves headroom to 3.28 V, so a fault current up to **32.8 A is measured rather than clipped** — which is what an overcurrent check needs. Offset is worth 30 mA of injector current (0.17 % of peak) and 3 mA of metering current (0.44 % of setpoint). Bandwidth resolves the 38 µs ramp to peak with about 0.45 µs of lag.

Supplied from 3V3_MCU, which makes the output ADC-safe by construction. It cannot swing above VS − 0.02 V, so these three channels need no 3.3 V zener at the MCU pin — unlike every channel on `sensors_analog`, which arrives from the harness.

This closes `MU-ISENSE`'s open half. That tag specified the 50 mΩ shunt and said "the amplifier below is not optional"; this is the amplifier. `IN-` goes to GND, and the tag's layout requirement stands: a Kelvin tap at the shunt's own ground pad, not a via into the pour.

`AUIRS2181STR` — chosen because it has *no* interlock

Infineon (International Rectifier) AUIRS2181(4)S, datasheet dated 10 January 2014, still hosted by Infineon. Retrieved and text-extracted this session. **Its first listed typical application is "Piezo / common rail Injection."** Confirm lifecycle status at sourcing — the datasheet is old even though the part is still listed.

Offset	VB **625 V** absolute maximum; VS operational **−5 V to +600 V**
Supply	VCC 10–20 V; UVLO+ 8.0 / 8.9 / **9.8 V** max
Logic	VIH 2.5 V min, VIL 0.8 V max — direct from a 3.3 V MCU pin
Output	1.9 A / 2.3 A typ, **1.4 A / 1.8 A min** short-circuit at 15 V
Delay	160 / 200 ns typ, 270 / 330 ns max
Interlock	**None** on 2181/21814 (the 2183/2184 have it)
Package	SOIC-8, AEC-Q100

Three properties chose it, each against a part that failed on it:

Property	This part	Why it matters	Failed it
No cross-conduction interlock	none	In a half bridge, HO and LO together is shoot-through and every driver prevents it. Here the "high" and "low" switches are in **series with the coil** and must **both** be on for it to conduct at all	UCC27282-Q1, UCC27712-Q1 — could not fire an injector
Offset rating	625 V abs max, **5.4×** margin	Bank node ~104.1 V worst case (41 µF reservoir), VB 12 V above it = 116.2 V	UCC27211A-Q1 — HB 120 V abs max, **3.8 V** of headroom

Property	This part	Why it matters	Failed it
Negative VS	operational to −5 V, clears by **3.7 V**	`D_fw` drops the bank node 1.0–1.3 V below ground on **every** hold off-time	UCC27211A-Q1 — HS DC min −1 V, **violated in normal operation**

The first is the one that is not a rating, and the one nobody searching a parametric table would filter for: the familiar reflex is that a half-bridge driver *without* interlock is the less safe choice. In this circuit interlock is not a safety feature, it is a failure to function.

Five in use. Four on `injector` — each package pairs one high side with one low side, by package and not by circuit; UB2's low side is spare. One on `metering_egr`, low side only, so the board carries one gate-driver part number instead of two.

The kill path moves to `HIN`. The part has no shutdown pin, and a ground-referenced FET cannot short a gate floating 100 V up. `HIN` *is* ground-referenced, so `supervisor.cir`'s kill FET goes there behind a 1 kΩ from the MCU. It is slower than the low-side gate clamp — 330 ns maximum driver turn-off against 23 ns — and that is acceptable because it is not the path the fail-safe argument rests on: the low sides are in **series** with every coil, and killing any one stops that cylinder.

Checked by `boost_converter`'s rating checks in `run_sim.py`.

`GATE-RAIL` — 12V_GATE, a discrete follower off the boost rail

The board had no rail between 5 V and 40 V, and the drivers need 10–20 V. **Its source is the boost rail, and the reason is cranking:** a regulator from `VBAT_PROT` cannot make 12 V from a 6–9 V cranking battery, and a gate rail that collapses during cranking means no injection exactly when the engine needs it. `BOOST_100V` is the one rail already designed to hold up from 6 V.

Not the LM5164 already on the board — its VIN absolute maximum is **100 V** (TI datasheet §5.1), and this rail sits at ~102 V.

Part	Value	Why
Bias	2 × 47 kΩ	87 V split so neither resistor carries all of it; 0.87 mA
Reference	**BZX84-C13-Q**	13 V − Vbe − limit drop = **12.07 V** at 5 mA; 11.35–13.55 V across tolerance and temperature
Collector	**2 × 3.3 kΩ, 2512**	Takes 33 V of the drop at 5 mA; 0.45 W each into a short
Pass	**PZTA42-Q**, SOT223	300 V, AEC-Q101. **0.27 W** normal, 0.26 W shorted → Tj 133 °C at a 105 °C board
Limit	**PMBT3904-Q + 56 Ω**	0.65 V / 56 Ω = **11.6 mA**
Output	10 µF 25 V	Holds through each driver's gate-charge pulse

Load is about 3 mA (five drivers' quiescent current plus gate charge), designed for 5 mA. **12V_GATE clears the drivers' UVLO+ ceiling by 2.27 V.** Standing load on the boost is now 9.65 mA — bleed, gate rail and divider — of its 50 mA, and the peak-phase draw still recovers in 8.48 ms against 26.67 ms between cylinders.

Why the collector resistors. The design first called for a DPAK NPN rated ≥160 V. No such datasheet could be retrieved: onsemi and Diodes both refused the download. The part that was retrieved, PZTA42-Q, is 104 K/W in SOT223, so it may dissipate only 0.43 W at a 105 °C board. The pass transistor alone would carry 0.44 W normally and 1.16 W into a short. Putting 6.6 kΩ in the collector moves half the power into two 2512 resistors. The cost is headroom: 12V_GATE now needs the boost rail above 46.5 V, which is half of its worst droop. **The 105 °C board ambient is an assumption** — no ambient is stated anywhere in the spec.

Its failure runs the safe way. If the boost stops, 12V_GATE collapses, every driver drops into UVLO and holds its outputs low, and every gate on the board turns off.

`TPS40210QDGQRQ1` — chosen on an off-time, not on an input range

TI SLVS861F, August 2008, revised June 2020. Retrieved and text-extracted this session.

Input	4.5 V–52 V, **absolute maximum 52 V** (Table 6.1)
Reference	700 mV, 686–714 mV over −40/+125 °C (Table 6.5)
Off-time floor	**tOFF(min) = 170 ns typ, 200 ns max** (Table 6.6)
UVLO	4.25 V typ, **4.5 V max** turn-on (Table 6.5)
Supply current	1.5 mA typ, **2.5 mA max**, not switching
Internal regulator	BP = 8 V typ (7–9 V), 0–15 mA
Gate driver	400 mA source / sink
Package	HVSSOP-10 PowerPAD (DGQ), 3 × 3 mm, AEC-Q100 grade 1

The number that chose it is the off-time, and the reasoning runs backwards from the usual. A boost from a 6 V cranking dip to 100 V needs

```
D = 1 - 6 / (100 + 0.9) = 94.05 %
```

and a controller's minimum off-time is a floor **in time**, not in duty. At 150 kHz, 200 ns is 3 % of the period, so `Dmax` = 97.0 % and the cranking corner clears by **2.95 points**. At 2.2 MHz — where most wide-input boost controllers in this class sit, and where the magnetics would be far smaller — the same 200 ns is 44 % of the period and `Dmax` falls to **56 %**, below even the 86.6 % the rail needs at a *nominal* 13.5 V battery.

So "higher switching frequency, smaller inductor" is the wrong optimisation here, and the part that looks less modern is the one that works.

THE RATING THAT DOES NOT FIT — the third time on this board. VDD's absolute maximum is **52 V**; `VBAT_PROT` reaches **73.3 V** for about 50 µs on ISO 7637-2 pulse 2a. Over by **1.41×**.

This is the same shape as DRV8873-Q1's 40 V `VM`, which took that part off the EGR bridge — and it is solvable here for a reason that did not apply there. A motor bridge's `VM` carries the motor current; this pin carries 2.5 mA plus gate charge. So it gets its own clamp:

`BOOST-VDD-CLAMP` — 220 Ω, 1 W, and a BZG03C36-HM3

Pulse 2a, 73.3 V	121 mA through the zener for 50 µs; **VDD 46.6 V** worst case against 52 V abs max (1.12×)
Load dump, 40 V for 400 ms	33 mA at the zener's cold minimum (32.7 V), **1.09 W** for 400 ms. The 1.25 W rating is steady-state at 25 °C, so this relies on thermal mass; not yet checked against a Zth curve
Cranking, 6 V in	2.5 mA + 8.7 nC × 150 kHz = 3.8 mA, dropping 0.84 V → **VDD = 5.16 V**
against UVLO turn-on max 4.5 V	**0.66 V of margin**; 0.125 V even at the 25 nC gate-charge ceiling

The worst case is taken as VZ max (38 V at 10 mA), heated to 125 °C at the +0.11 %/K maximum tempco. On top of that sits the 40 Ω maximum dynamic impedance, applied from 10 mA up to the pulse current. That last figure is a small-signal value at 10 mA, and dynamic resistance falls with current, so it is a bound, not an estimate.

What was drawn first, and why it was wrong. The first design was 47 Ω and "a 43 V zener", and it was checked as an ideal 43.0 V source. The real part in that class, BZG03C43-M, is 40–46 V at 10 mA, up to +0.12 %/K, with a 45 Ω impedance. With 47 Ω in series, a quarter of an amp flows during the pulse. At 125 °C that puts **61.9 V** on a 52 V pin. The value was also chosen so the clamp would stay out of conduction during the 40 V load dump, but its tolerance band starts at 40.0 V. Found on 22 September 2026, when the zener was chosen and its datasheet read. The old design is kept as a pinned check.

The fix is in the resistor. The chosen boost switch, DMN15H310SK3, has a small gate charge, which lowers the pin's current enough to allow 220 Ω. That cuts the pulse current by a factor of five and limits load-dump conduction to 33 mA. Once the dump current is limited, a lower zener is safe, and it gives back the margin the tempco takes. 39 V was tried first and gave 49.4 V. That is inside 52 V but only 1.05×, below the 1.09× the `transient_clamp` rail table requires of every part on this rail, and the table failed it.

The power stage is untouched by any of this — the inductor and FET see the rail directly, and are 150 V class for it, the same class the injector switches already carry.

Checked by `boost_converter.cir`'s rating checks in `run_sim.py` — not with a switching model, which that block deliberately does not have, but as executable arithmetic on the datasheet's own numbers. Same treatment `transient_clamp.cir` gives the 73.3 V rail.

What is NOT settled by a datasheet: the loop compensation. `Rcomp` 10 kΩ and `Ccomp` 6.8 nF put the error amplifier's zero at 2.3 kHz, below the right-half-plane zero a boost puts at `(1-D)² · Rload / (2π·L)` = 17.3 kHz at 13.5 V in — and that zero **moves with duty cycle**, so the cranking corner is the hard one. These are a starting point. A current-mode boost's loop depends on the inductor's real DCR, the capacitor's real ESR and the FET's real switching behaviour; it gets measured on the bench.

`LM5164DDAR` — and the 60 V part it replaces

Research memo 07 §4 selected **TPS54360B-Q1**, reasoning that a 60 V part beat a 42 V LM5175-Q1 "for load-dump margin". That reasoning was correct when it was written. It is not correct now, and the change came from the input-protection work above:

- The TVS standoff had to rise to 43 V so the part stops conducting during a *normal* clamped load dump.
- A higher standoff clamps higher. Pulse 2a now clamps at **73.3 V**.
- `transient_clamp.cir` carries a **passing** check that says so in as many words: *"60 V-class parts are NO LONGER viable downstream."*

A 60 V buck on this input is a part that fails the first ISO 7637-2 pulse 2a event. The LM5164 was already the part every rating check in the suite was written against — its −0.3 V / 100 V VIN limits are what `negative_pulses` and `transient_clamp` check — but nothing had ever reconciled the BOM table with it. **This is what a contradiction between two documents looks like when only one of them is executable.**

Operating range 6–100 V covers the spec's own 6–40 V requirement, with the 6 V end matching the cranking dip the input range was built around. 1 A against an estimated 400–500 mA board load.

Configuration values, all derived from the datasheet rather than picked:

Part	Value	From
`RON`	31.6 kΩ	The tON table in §5.5 is linear in RON/VIN: at 12 V, 25 kΩ gives 830 ns, so K = 398 ns·V/kΩ. For 400 kHz at 5 V out, RON = tON·VIN/K is 31.4 kΩ at **both** 13.5 V and 40 V — the VIN feedforward is what holds the frequency fixed. E96 value 31.6 kΩ.
FB divider	38.3 kΩ / 12.1 kΩ	VREF is 1.2 V (1.181–1.218, ±1.5%), so 5 V needs 1 + R1/R2 = 4.167.

Part	Value	From
EN/UVLO divider	75 kΩ / 24.9 kΩ	Enable rises at 1.5 V typ; 0.249 ratio puts turn-on at 6.02 V, the spec's cranking-dip floor. The pin is rated to 100 V, so the divider programs the threshold rather than protecting the pin.
BST cap	2.2 nF 50 V X7R	Not a range. The datasheet specifies the part: *"a high-quality 2.2nF 50V X7R ceramic capacitor between BST and SW"*.
`L` / `Cout`	33 µH / 47 µF, 10 mΩ ESR	`buck_preregulator.cir`, sized at the 40 V worst case rather than nominal. The ESR is part of the value — it sets the ripple.

The RON figure is worth one more line: it independently reproduces `buck_preregulator.cir`'s own gate timings, 925 ns at 13.5 V and 312.5 ns at 40 V. The netlist got those from D = Vout/Vin; the datasheet gets them from RON. They agree.

`TLV76733QWDRBRQ1` — accuracy as a requirement

Memo 07 §6 named `TLV1117-33QDCYRQ1`, explicitly *"class pricing, not fetched"*. Choosing the supervisor turned 3V3 accuracy into a hard constraint: outside **3.143–3.459 V** the board resets itself. TLV767-Q1 is **1% over load and temperature** — 3.267–3.333 V, about 125 mV of margin at each end. 1 A, 2.5–16 V in, AEC-Q100, VSON-8.

The fixed-output version is the one specified. Its pin 3 is `SNS` where the adjustable version has `FB`, and they are not interchangeable. `SNS` must not float; it is tied to `OUT`.

`TPS3850G33DRCT`

See `docs/pinmap.md` §1.8 and `hw/gen_symbols.py`. The variant letter is the decision: `G` sets thresholds at ±4% of nominal and `H` at ±7%, and with the part's ±0.8% accuracy that is a worst-case undervoltage trip of **3.143 V** against **3.044 V**. The supervisor has to assert before the S32K148's own LVD (3.0 V max), so G33 clears it by 143 mV and H33 by 44 mV — close enough that the two could fire in either order, and a supervisor that might lose the race to the thing it supervises is not doing its job. The cost is the tighter rail window above.

MCU

NXP S32K148, `FS32K148HAT0MLQT` — 144-pin LQFP, 80 MHz, −40…+125 °C.

Confirmed in research memo 05 against the S32K1xx data sheet Rev. 15 with page-level citations. The full reasoning is there; what matters for the BOM:

- The **144-pin** package is headroom against unknown U1 (50 of the OEM ECU's 94 pins are still unresolved), not a requirement of the confirmed channel budget, which fits inside a 100-pin part.

- **LQFP over MAPBGA**: identical instance counts — CAN/ADC/FTM/I²C are silicon-level, not package-level — and 0.5 mm pitch is inspectable by eye or AOI, where a BGA needs X-ray or destructive inspection. A real cost for a small team doing early assembly.
- **80 MHz `H` grade, not 112 MHz `U`**: the `U` speed grade is *not valid* at the 125 °C `M` temperature grade, per the datasheet's own ordering-table footnote. Memo 05 §6 shows the extra clock is not needed to close the timing budget, so there is no reason to trade thermal margin for headroom the design does not use. This corrects the spec §5 framing, which cites 112 MHz.
- In stock at DigiKey's India storefront at single-unit quantities. Not a lead-time part. The S32K146 fallback was checked and is *not* needed — it is the one that is out of stock, with 12–52 week lead times.

Still to check before footprint lock: package drawing 98ASS23177W (SOT486-2) has not been independently fetched. The PCB roadmap's warning stands — a hand-sourced footprint is the most likely place to introduce a silent error.